

The Bayesian Fair Value of Carbon: A Probabilistic Framework for Risk-Adjusted Climate Finance and Long-Term Solvency

Abstract

The problem: The Voluntary Carbon Market (VCM) is paralyzed by the current “fungibility crisis”

The VCM is failing to scale, in large part due to pervasive quality concerns and lack of trust that carbon credits truly achieve their aims. For decades, the market has assumed that a “ton is a ton”, regardless of the fact that projects differ greatly in their risk profiles, permanence horizons, and broader social/ecological impacts. Operating under the “**nominal ton fallacy**” has let low quality credits linger on the market, left registries insolvent in the face of reversals, fed into greenwashing concerns from buyers, and prevented capital from efficiently flowing to truly impactful climate solutions.

The solution: A Bayesian Fair Value (BFV) Framework

This paper introduces the **Bayesian Fair Value (BFV) Framework**, an actuarial accounting system that quantifies the probability of every credit achieving realized climate impact. Rather than treating carbon credits as a fixed ton with varying quality, the BFV treats them as **probability-weighted assets**. Every credit is quantified as a “fair value” unit that represents its realized climate impact relative to one nominal ton, integrating four core variables into a single metric: **Execution Risk, Scientific Uncertainty, Permanence, and Broader Co-benefits**.

Key innovations of the BFV Framework

The BFV “Engine” introduces three new mechanisms, aimed at solving some pervasive issues that have limited growth of the carbon market:

- **A social and nature benefits scalar:** Rather than treating broader benefits and risks as a separate evaluation, the BFV Framework integrates social and ecological externalities into the ‘fair value’ ton. This allows for these targets to be directly integrated into optimized portfolios, alongside climate targets.
- **Dynamic uncertainty banking:** The probabilistic accounting approach replaces static buffer pools with a dynamic accounting model that incentivizes developers to lower scientific and execution risks while protecting both sides of the market from stochastic events.
- **A Universal Climate Ledger:** BFV fair value tons can be calculated for any climate investment, providing its realized climate impact is estimated probabilistically. This means the core logic can be extended to include decarbonization, sovereign NDCs, and other climate actions.

The Outcome: Actuarial Solvency

The BFV Framework is not meant to replace current climate integrity or corporate climate action standards. Rather, it is an accounting layer that integrates with our collective vision of how to achieve an impactful, just, and equitable climate transition. By providing the mathematical framework to calculate the **Cost of Certainty**, the BFV Framework lays out the route to move from accounting nominal tons to accounting the true, realized impact of climate investments.

The Problem: The crisis of fungibility

The VCM suffers from a **fundamental fungibility crisis**.

Since its inception, the market has operated on the 'Nominal Ton' fallacy—the implicit assumption that 'a ton is a ton.' In reality, carbon credits possess **divergent risk profiles, disparate permanence horizons, and asymmetric co-benefit utilities**. This forced fungibility obscures the true cost of certainty, creating the systemic integrity gaps that now paralyze global climate finance.

Overview of the current market

Some progress has already been made. Since 2020 or so, three trends have helped supplant the view that 'a ton is a ton', and have led to greater recognition that individual carbon credits are not strictly fungible:

- **The “flight to quality”**: High profile media criticism, first of over-crediting in REDD+ programs, raised awareness that the risk profiles of carbon credits can vary dramatically from project to project.
- **Focus on net zero**: The emergence of net zero targets as the main motivator of many credit buyers has split the market based on credit durability, with high profile policymakers such as SBTi shifting demand towards permanent carbon storage.
- **Rise of nature and other co-benefit targets**: At the same time, more organizations are focusing on projects with strong nature and social impact claims, further complicating the market, particularly for nature-based credits.

These are all positive developments, but in the fractured voluntary carbon market, the mechanisms that have emerged have sometimes caused as much confusion and mistrust as they have presented solutions. The current status quo is a patchwork of measures enacted by registries, 3rd party policymakers, credit investors, and ratings agencies, aimed at addressing some of these gaps, but not all of them.

These interventions—including flat rate buffer pools, static risk adjustments, and quality ratings—are rooted in the idea that 'a ton is a ton', or at least that carbon credits can be made fungible with rubrics and simple math. The result is a market that remains incapable of pricing the '**cost of certainty**.' This leaves registries insolvent in the face of long-term reversals, buyers exposed to 'greenwashing' litigation, and policymakers unable to direct capital where it is mathematically most effective.

To resolve the fungibility crisis, we must move beyond the 'Nominal Ton' and establish a unified, probabilistic accounting of the risks and benefits arising from the full range of climate solutions available on the current market.

The Solution: A Bayesian Fair Value Framework for Climate, Nature, and Social Impact targets

To move past the **Nominal Ton**, we must stop treating climate benefits as static commodities and instead regard them as **Probability-Weighted Climate Derivatives**.

The **Bayesian Fair Value (BFV) Framework** provides the actuarial infrastructure to accomplish this. Rather than issuing qualitative ratings, the BFV Framework calculates the **Fair Value** of carbon claims based on their full posterior distribution of risks and benefits. It provides the mathematical strategy required to maintain the **long-term solvency** of portfolios containing heterogeneous project types—from nature-based removals to engineered sequestration.

Critically, the BFV Framework does not disrupt existing climate finance workflows. It introduces a **probabilistic accounting layer** that sits atop current standards, replacing subjective claims of "credit quality" with a rigorous, data-driven "cost of certainty."

Mathematical overview of the BFV Framework

The BFV Framework at a glance

The BFV Framework replaces “nominal” tons with a fair value accounting unit. We define the Bayesian Fair Value (BFV) as the expected impact of a climate claim, incorporating various underlying risks:

$$BFV = (P \times E[\theta]) + \Omega$$

Where:

- **P (The nominal unit)**: For the purposes of this framework, *P* is set to 1.0, representing one “nominal ton” of CO₂E. By standardizing *P* to one ton, the BFV represents the **risk adjusted realized climate impact** of each credit.
- **E[θ] (the expected delivery coefficient)**: This is the mean of the joint posterior distribution of risk. It represents the probability that each ton of carbon is sequestered, properly measured, and maintained in terrestrial storage over time.
- **Ω (the co-benefits scalar)**: This is a value that represents the benefit or debt to nature, social impacts, and other co-benefits associated with each credit. In the BFV Framework we will define this value in “carbon equivalent units”, allowing for direct aggregation of co-benefits into the BFV claim.

Moving from “nominal tons” to realized climate impact

One of the chief benefits of the BFV Framework, particularly when defined based on realized climate impact as above, is that it lets us directly assess the solvency of each credit given the data, assumptions, and other ancillary information we have available.

For example if a project has an $E[\theta]$ of 0.7 and an Ω of 0.1, its BFV value is 0.8. From a registry or portfolio management perspective, this tells us that each ‘nominal ton’ is only 80% solvent. To maintain a 1:1 integrity standard, we must use this coefficient to dictate issuance and reserve requirements—ensuring that for every ton promised to the atmosphere, a full “Fair Value Ton” is mathematically accounted for.

Rules of the “BFV Engine”

The BFV Framework is **methodologically flexible by design**. The “Bayesian” in its name refers to how it applies Bayesian logic to dynamically manage credit risk:

- **Prior elucidation:** At project initiation, derive prior distributions of key components based on initial data and domain expertise.
- **Data collection & modeling:** Use these priors to identify knowledge gaps, then develop modeling and data collection approaches. These are integrated with priors to develop posterior expectations of BFV.
- **Dynamic updating:** Continually update with the latest data on a periodic basis; the previous cycle’s posterior becomes the prior of the next iteration of BFV.

Using a fully Bayesian modeling framework is the recommended approach, but it is not strictly required. The expected delivery coefficient $E[\theta]$ can be modeled with any approach, so long as it respects three “rules” for properly calculating BFV.

Rule 1: BFV inputs must be expressed as probability distributions.

At its core, the BFV framework is a method for quantifying climate benefits as probability-weighted assets. It does not accept binary claims of “integrity”, but instead treats carbon credits as probability distributions.

Action: To express BFV as a probability distribution, the joint posterior distribution of execution, science, and permanence risk $E[\theta]$ must also be the product of a set of probability distributions. The co-benefit scalar Ω is not strictly required to be a probability distribution but will ideally be expressed as one also.

Result: A credit with an 80% chance of delivering a full ton of carbon storage—the result of $E[\theta]$ —is issued 0.8 BFV units. This eliminates the ‘nominal ton’ fallacy from the very beginning of the crediting process.

Rule 2: Projects with lifespans shorter than the permanence benchmark must be rolled over.

In the BFV framework temporary carbon storage, such as from nature-based solutions, function as “limited-term notes”. The engine treats their expiration not as a failure, but as a scheduled maturity.

Action: For any asset with $P(\text{Permanence}) < 1.0$ the BFV engine calculates a replacement cost—the mathematical requirement of rolling over temporary storage into another asset before the end of its crediting period.

Result: This turns a static purchase into an **Actuarially Solvent Portfolio**. The engine ensures that as one “Nature Note” expires, a successor (either another NbS note or a permanent anchor) is already mathematically accounted for.

Rule 3: Portfolio re-balancing is dynamic and based on uncertainty of its assets.

To remain solvent, portfolios using the BFV framework must maintain solvency in real-time, not merely as a historical condition.

Action: As new information (refined monitoring data, changes in geopolitical status, improved carbon science, etc.) enters the model, the **priors are updated**.

Result: The goal is to maintain as close to a 1:1 ratio between BFV units and nominal tons as possible. With wider uncertainty intervals, this gap grows. Rather than static buffer pool contributions, this approach induces a **dynamic uncertainty reserve** that tracks the risk profile of the portfolio.

In the next section, we will go deeper into the components of a “fair value ton”. We will also look at some potential model structures that are good fits with BFV. However, as outlined above, these should be taken as best practice recommendations rather than strict requirements.

The decomposition of a “fair value ton”

The expected value coefficient ($E[\theta]$) is the heart of the BFV framework. We define it as the joint posterior distribution of three risk vectors:

$$E[\theta] = P(\text{Execution}) \times P(\text{Science}) \times P(\text{Permanence})$$

In this section, we will first provide a conceptual overview of each component, and outline proposed robust modeling approaches for each.

Execution risk (P(E)): The Operational Prior

Definition: The probability that a project is successfully implemented and maintained, following its methodology and design. This includes various operational tasks such as earning permits, executing planting schedules, raising funding, meeting planned issuances, etc., etc. The correct set of benchmarks for estimating $P(E)$ will vary project to project.

Box 1: Modeling $P(E)$ as a dynamic posterior

Execution risk $P(E)$ can be quantified using a Beta-Binomial model such as:

1. **Operational prior:** At project initiation an initial probability distribution $P(E) \sim \text{Beta}(\alpha, \beta)$ is set based on developer track record, jurisdictional risk, and project complexity. For example a new developer operating in a somewhat risky jurisdiction might receive a $\text{Beta}(2, 2)$ prior, effectively saying we believe there is a 50:50 chance of the project succeeding.
2. **Milestone integration:** The project establishes a schedule of performance-based milestones such as establishing legal land tenure, securing funding, or executing planting operations.
3. **Performance-based updating:** As milestones are achieved the BFV Engine integrates these results and updates $P(E)$ accordingly. Successes shift the distribution towards 1, increasing the BFV of the project. Delays or failures reduce $P(E)$, increasing the cost of certainty and signaling the project is at risk.

This approach ensures $P(E)$ is an earned metric, driven by successes on the ground. This MVP model can be extended to introduce additional components or covariates on a case by case basis.

Science risk ($P(S)$): The Climate Impact Claim

Definition: The probability that the carbon accounting framework—including the methodology as well as modeling decisions, etc.—reflect reality and provide accurate predictions of carbon sequestration produced by the project. This includes risk related to additionality (whether the project has a sound scientific approach to producing new carbon storage) as well as estimation of baseline and project scenarios (model error, measurement error, etc.).

Box 2: Best practices for estimating science risk $P(S)$

The best practice for estimating $P(S)$ is to fit a Bayesian hierarchical model that allows us to estimate the true climate impact while accounting for various sources of uncertainty from monitoring data, modeling assumptions, and stochastic events.

$$y \mid \theta, \sigma \sim f(\theta, \text{data})$$

Where y is observed measurements (i.e., forest carbon measured on plots, CO_2 concentrations from lab samples, etc.), θ is the true climate impact, and σ is the model variance.

The generic likelihood function $f(\cdot)$ could be a Gaussian, Poisson, or any other statistical distribution, depending on the structure of the project data. The true climate impact θ will

likewise vary. For a forest carbon project it may be a Chapman-Richards growth curve, while a DAC project may use a probabilistic model of the CO₂ absorption and sequestration process. A basic model can be expanded to accommodate other components such as leakage. Using a hierarchical structure with global hyper-priors allows full propagation of uncertainty into the estimation of $P(S)$.

Permanence risk ($P(P)$): The Net Zero Anchor

Definition: The probability that sequestered carbon will remain out of the atmosphere for a 1000-year “geological scale” time horizon, the accepted minimum for “permanent” carbon removal to meet net zero claims.

Most current frameworks treat permanence as variable, and assess risk that the project will fail to meet this claim. However, allowing the target time horizon to vary fails to establish equivalence among different types of carbon projects, and disadvantages the use of temporary (nature-based) credits in portfolios that address net zero claims. By fixing the time horizon to 1,000 years, we are establishing a universal denominator that can apply to any credit.

Box 3: A proposed model for estimating $P(P)$

A **Bayesian survival model**, such as a Weibull or Cox Proportional Hazards model, makes it convenient to calculate the “permanence discount” for our fixed 1,000 year timeframe.

This approach involves designing a **hazard function** that will predict the probability of a reversal at any time t . The shape of these functions will vary based on the type of project. For nature-based projects, the probability of reversal typically increases over time. For engineered projects it may be flat or even decreasing, to reflect lowering storage risks as engineering hurdles are addressed. Hazard functions can also incorporate project-specific risk factors, such as regional climate resilience or political instability.

The co-benefits scalar: Optimizing for nature, social, and other co-benefits or risks

Definition: The co-benefits scalar (Ω) is a quantitative value that represents verified externalities (both positive and negative) of the carbon project, expressed in “carbon equivalent units”. These co-benefits could be enumerated in any way, but mostly will include those frequently addressed by corporate ESG programs: biodiversity, social equity, and water security.

Crucially, Ω is symmetric, meaning it can both reward and penalize projects. A high-quality ecological restoration project that effectively engages local communities, or a biochar project that sources “biodiversity friendly” feedstocks, would have positive values of Ω . Conversely a

planting of non-native species, or a DAC facility that disrupts local water supplies, could have negative Ω .

In essence, Ω is a “safety valve” that penalizes projects with very strong climate claims if they create other risks, and rewards riskier projects when they have obvious benefits. As we’ll see in the examples below, Ω is critical for promoting “nature positive” net zero.

Data Inputs:

- **Biodiversity:** Species richness indices, habitat connectivity metrics, or eDNA monitoring results.
- **Social:** Verified community benefit-sharing ratios, local employment data, and alignment with UN Sustainable Development Goals (SDGs).
- **Water:** Watershed restoration metrics or groundwater recharge data.

The Bayesian View: We treat Ω as a multi-attribute utility function. Much like $E[\Theta]$, we define “sub-models” of biodiversity, social, and water impacts, then update the overall scalar given available data and domain expertise.

Uncertainty Banking: Maintaining solvency through time

The BFV is more than an accounting standard; it is a blueprint for a **dynamic solvency stack** capable of maintaining atmospheric integrity through time. The framework moves away from static, "one-size-fits-all" buffer pools and toward a three-tiered capital structure designed to manage risk at the project, portfolio, and systemic levels:

The BFV Solvency Stack: A three-layered approach to maintaining atmospheric solvency

Layer	Financial Tranche	Risk Type	Mechanism	Recovery Trigger
1. Project Escrow	Equity (First-Loss)	Endogenous: (Execution, Science, Specific Reversal)	Non-circulating credits withheld at issuance based on the (1 - BFV) Integrity Gap.	Performance Release: Improved data or age-driven stability "unlocks" credits for sale.
2. Central Repository	Mezzanine (Liquidity)	Idiosyncratic: (Localized Stochastic Shocks)	A diversified "Mutual Fund" of credits/cash funded by a % fee on every BFV transaction.	Liquidity Bridge: Issues "low-carbon-interest" loans to projects that exhaust Layer 1.
3. Insurance Backstop	Senior (Systemic)	Exogenous: (Black Swans, Regional Catastrophe)	External underwriter policy providing "Excess of Loss" capital.	Capital Infusion: Cash/Credit payout to recapitalize the Repository and maintain registry-wide solvency.

The keys to this strategy is that it is **performance-based**, particularly for endogenous risks (i.e., those most closely related to project design) and that it **aligns different risk management strategies for different types of risks**. In a traditional buffer pool, the same approach is used for any type of loss (overcrediting) or reversal. The BFV Framework recognizes multiple risk management strategies, and aligns accordingly.

Layer 1: Project-specific escrow (Addressing execution, science, and other directly manageable risks)

The project escrow is the first line of defense and it collateralizes **internal risks** to the project's design and execution. These are things developers can directly control, such as deployment of high quality models and staying on track to reach project milestones.

- **How it works:** Upon issuance, the “integrity gap” is calculated as $1 - \text{BFV}$. This is the proportion of credits that will be held in escrow to guard against endogenous risks.
- **Example:** A project generating 1,000 nominal tons receives a BFV of 0.75. 750 tons are available to issue immediately while the remaining 250 are held in escrow.
- **The Logic:** Unlike a traditional buffer pool these escrow accounts are owned by the developer but managed by the registry and kept out of circulation. By using the integrity gap and dynamic updating, this approach provides a **financial incentive** to developers to improve the quality of projects. Methodological changes that increase $P(E)$, $P(S)$, and/or $P(P)$ —improving MRV approaches, for example—will reduce the number of credits held in escrow.

Layer 2: The Central Repository (Managing localized stochastic shocks)

To maintain liquidity while avoiding over-penalizing developers for stochastic events, the BFV Framework establishes a **central repository** of credits that cover timebound shortfalls.

- **How it works:** When a project experiences a localized loss of credits due to stochastic events, the registry issues a “**bridge loan**” of comparable credits. This ensures purchased units remain solvent, while allowing developers to restock their escrow account on a planned time horizon.
- **Example:** A BECCS project loses 500 credits of their next issuance when a sudden policy shift shuts their main feedstock supplier. They only had 250 credits in escrow. The bank issues a bridge of 250 credits from a similar project, and the BECCS developer plans to increase their escrow contributions over a 10-year period to make up the shortfall.
- **The Logic:** Carbon projects are exposed to risks developers cannot control, such as changes in jurisdictional governance and natural disturbances. The central repository avoids over-penalizing developers and putting projects at risk, while still maintaining long-term solvency. Credits in the repository are purchased using funds from participating developer and buyer fees.

Layer 3: The Insurance Backstop (Managing systemic failure)

The final layer is an insurance backstop meant to protect against “black swan events”, such as regional catastrophes that threaten the portfolio’s solvency

- **How it works:** Since the portfolio is significantly de-risked through layers 1 and 2, the registry presents a strong profile to external underwriters. This allows for affordable, large-scale excess of loss insurance.
- **The Logic:** In the event of a catastrophic failure, the insurance payout provides the capital necessary for the registry to purchase replacement credits on the open market, ensuring that the total “Fair Value” in circulation never exceeds the physical carbon in the ground.

Creating a positive incentive loop while minimizing reputational risks

Compared to traditional buffer pools, this approach presents two main advantages:

- **The BFV Framework incentivizes project improvement:** By managing endogenous risks through an escrow account, project developers are incentivized to seek improvements that will allow more of their credits to flow to the market. Rather than paying a set rate to the registry regardless, it creates a clear **incentive** to increase project quality and narrow the “integrity gap”. The registry is likewise incentivized to diversify the project portfolio, and to maintain integrity standards that will lower exposure to idiosyncratic and systemic risks.
- **It reduces reputational risks for buyers:** Layers 2 & 3 ensure that buyer portfolios remain solvent even in the event of large reversal risks. This alleviates a significant reputational concern for buyers, particularly for ‘riskier’ project categories such as certain nature-based projects and emerging CDR technologies.

None of this is to say that managing a BFV registry is without its challenges. The introduction of Layer 2 expands the role registries play in the current market, requiring them to originate and manage their own credit supply. The process of matching “like for like” credits to create liquidity bridges is also non-trivial.

But, it’s worth highlighting that this isn’t fundamentally different from how the market currently functions. In the current market the various processes—assessing credit integrity, issuing & verifying credits, and managing supply—are split among multiple firms whose goals don’t necessarily align. Further, many buyers are already managing reputational risk by simply not acting.

The BFV Framework doesn’t change the rules of the current carbon market. It simply provides an actuarial framework that clearly elucidates the risks, mitigates them to the greatest extent possible, and provides a system where developers, buyers, and policymakers are incentivized to make risk-informed decisions.

BFV in action: An example nature-based and CDR portfolio

To demonstrate the BFV in action, we will look at an example carbon credit portfolio and consider how three different actors—a registry, a corporate buyer, and a global policymaker—can utilize the framework to make risk-adjusted investment and management decisions.

The portfolio

Our example portfolio consists of 10,000 “nominal tons” split over four project archetypes (Table 1):

- **Industrial IFM:** A US-based timberland firm executing an improved forest management (IFM) project on managed pine plantations.
- **Mangrove restoration:** A small NGO managing a mangrove ecosystem restoration project in eastern Africa.
- **Industrial CDR:** A biochar project connected to large-scale sugarcane agriculture in Brazil.
- **Frontier tech:** An ocean alkalinity enhancement project in northern Europe.

Table 1: Portfolio distribution in “nominal tons”, or estimated climate impact prior to BFV risk adjustment

Project Archetype	Nominal Tons
Industrial IFM (US)	5,000
Mangrove (Small NGO)	2,000
Biochar (Industrial)	2,000
OAE (Frontier Tech)	1,000
PORTFOLIO TOTAL	10,000

Example 1: How a registry uses BFV to manage risk and maintain long-term portfolio solvency

For this example, we will assume the registry functions as a “central bank” that is capable of evaluating projects, calculating relevant BFV inputs, assigning nominal ton to BFV conversions, and managing the dynamic uncertainty bank.

Year 1: Intake

These four projects enter the registry’s credit book in the same cohort. Upon intake, the registry develops initial models of $P(E)$, $P(S)$, and $P(P)$, which allows for calculation of the joint posterior probability $E[\theta]$ (Table 2).

The two ‘industrial’ projects, IFM & biochar, receive high execution scores because they are embedded in ongoing industrial operations, in countries with low geopolitical risk. The mangrove

project receives the lowest $P(E)$ due to a high risk location and early stage, inexperienced developer. The OAE project falls in between. While it is well funded and located, the developer will have to navigate a lengthy permitting process to get permission to operate. The $P(S)$ scores follow a similar distribution: high for biochar and IFM, reflecting that these are well established pathways and the developers provided sound datasets, while lower for the more challenging modeling exercises (mangrove restoration and OAE). For $P(P)$ the mangrove project receives the lowest score, driven by risks of sea level rise, and the two CDR projects receive scores very close to 1.0, reflecting that they effectively create permanent carbon storage.

Note that calculating $E[\Theta]$ as the joint posterior distribution means that the conversion to ‘nominal ton’ is driven by the “weakest link”, thereby preventing any one attribute from having a disproportional impact on $E[\Theta]$. Current net zero guidance, such as from SBTi, would value the OAE project above either nature-based project simply due to its potential permanence. The BFV recognizes this as a strength but also penalizes the operational and quantification challenges that are unavoidable in this pathway.

Table 2: Calculated BFV inputs for the portfolio.

Project Archetype	Nominal Tons	$P(E)$	$P(S)$	$P(P)$	$E[\Theta]$
Industrial IFM	5,000	0.95	0.9	0.75	0.64
Small NGO Mangrove	2,000	0.55	0.65	0.6	0.21
Industrial Biochar	2,000	0.95	0.98	1	0.93
Frontier OAE	1,000	0.7	0.45	1	0.32
PORTFOLIO TOTAL	10,000	--	--	--	--

After quantifying $E[\Theta]$ for each project, a second set of analyses is conducted to assign the co-benefits scalar Ω . As expected the mangrove project carries a high, positive value of Ω because of its significant biodiversity and social impact value. The IFM & OAE projects receive small positive scores, reflecting limited co-benefits impact. The biochar project receives a deduction because it is unable to verify the source of its sugarcane feedstocks, and agriculture is a major driver of deforestation in the region.

With $E[\Theta]$ and Ω estimated for each project, we can now calculate the BFV Unit—the risk-adjusted fair value of one nominal ton from each project—and determine how many credits from each must be held in escrow.

Of the 10,000 nominal tons brought to the portfolio, 6,400 “fair value” units are available for immediate issuance, while the remainder are held to back the dynamic uncertainty bank.

Table 3: Year 1 calculation of BFV Units available for issuance and credits held in escrow.

Project Archetype	Nominal Tons	$E[\Theta]$	Ω	BFV Unit	Total BFV	Escrow
Industrial IFM	5,000	0.64	0.05	0.69	3,450	1,550
Small NGO Mangrove	2,000	0.21	0.40	0.61	1,220	780
Industrial Biochar	2,000	0.93	-0.25	0.68	1,360	640
Frontier OAE	1,000	0.32	0.05	0.37	370	630
PORTFOLIO TOTAL	10,000	--	--	0.640	6,400	3,600

Year 5: Dynamic updating: the “value release” and “margin call”

While the BFV framework is a convenient approach for risk-adjusting credits at a single point in time, its true utility comes from responsiveness to new data and information (the application of Bayesian logic to update our most recent priors, as discussed above).

Five years after initial intake, two significant events occur that require re-balancing of the uncertainty bank.

Event A: OAE project achieves a scientific breakthrough

The OAE project successfully published a peer-reviewed monitoring methodology, and has taken steps to deploy a sensor network that will field-validate their laboratory results.

- **The update:** $P(S)$ was estimated as 0.45 in year 1. After these advances, it **risks to 0.8**.
- **The result:** The BFV unit **increases from 0.37 to 0.61**.
- **Value release:** The registry **issues an additional 240 tons** which the developer can now sell on the market. The developer realizes a 65% increase in their supply without scaling the project—they gained by reducing the **cost of certainty**.

Event B: Wildfires impact the IFM project

In year 4, a severe wildfire impacts the IFM project. Post event monitoring reveals that 1,000 “nominal tons” of previously sequestered carbon have been lost to the atmosphere.

- **The write-down:** The registry recognizes the loss of 1,000 tons from the portfolio.
- **The solvency check:** Because the initial BFV of the project was 0.69, the 3,450 issued tons—all of which have since been purchased—were only backed by 69% of the physical forest in the project area.
- **Escrow absorption:** The registry cancels 1,000 tons from the IFM project’s escrow account, leaving a remaining balance of 550 tons.
- **The result**
 - **Buyer investments are protected:** Since the registry accounts for the loss, the 3,450 “fair value” credits that have already been issued and sold remain valid.

- **The credit bank remains solvent:** The developer provided collateral to guard against future “uncertainty debt”. In this case, it was enough to cover the shortfall created by the wildfire.
- **The developer commits to restocking the escrow account:** Since a fire is a stochastic event, the central bank releases 1,000 “like for like” credits as a “bridge” to keep the project solvent in the eyes of the market. The developer implements a fire management plan and is confident that they can mitigate fire-related reversals in the future. They will divert a greater percentage of new issuances (50% instead of 31%) to restock the escrow account, until the “uncertainty debt” is repaid.

Summary for the registry

BFV provides a clear, mathematical framework to value different carbon removal pathways as “fair value” assets. The dynamic uncertainty banking system induced by BFV’s probabilistic approach allows the bank to remain solvent while rewarding a developer for improving the quality of their project and protecting another from a stochastic reversal that was difficult to predict.

In this case, the escrow account was sufficient to protect against the reversal, and the bank is able to provide a low-interest bridge loan in the form of comparable credits while the developer restocks over a set time period.

Example 2: How a corporate buyer uses BFV to manage climate investment risks

A Fortune 500 company needs 10,000 ‘nominal tons’ to meet their performance benchmarks towards their net zero target. Under a traditional model they would simply buy 10,000 tons of credits. But, their sustainability office already knows that credits rarely represent one ton of actual carbon storage. Rather than hitting targets, they are shifting to **manage a carbon balance sheet**.

Initial acquisition

The buyer decides to purchase the entire 10,000 ton portfolio from example 1. However they recognize that, at least initially, they are only receiving **6,400 BFV units** towards their 10,000 ton goal.

- **The integrity gap:** The buyer needs to address a 3,600 credit shortfall. This is a sizable gap, but they recognize that had they purchased the “nominal tons” those 3,600 credits would represent a significant risk to their balance sheet.
- **The action:** The buyer wants to be “BFV solvent”. They contract additional credits from other, high-integrity projects to bridge the gap.
- **The value:** The buyer has naturally accounted for the risks inherent to the types of projects they’re purchasing. They can be confident their investment represents 10,000 tons of realized climate action, with room for upside as projects within the portfolio mature.

Year 5: The “value release” of the OAE project

In year 5, the buyer realizes some upside when the OAE project publishes a methodology and raises its P(S) score.

- **Asset appreciation:** The buyer previously owned 370 BFV units from this project. With the methodology change, this increases to 610 BFV units.
- **Result:** The “fair value” of the buyer’s portfolio has increased by 240 units. They could devote these credits to other targets, keep them and maintain their own internal buffer, or sell them to other corporations.

Year 5: The managed reversal (IFM project wildfire)

In a traditional market, a large-scale reversal presents a significant legal and reputational risk to buyers, particularly if the registry buffer pool is over-leveraged and can’t make up the deficit. In the BFV Framework, reversals are managed to avoid buyer exposure to the loss.

- **The backstop:** Because the registry maintained a layer 1 escrow based on the project’s BFV score of 0.69, the 3,450 units held by the buyer are still 100% solvent.
- **The write-down:** The reversal was absorbed by the developer’s escrow account, not by the buyer. Though the buyer does pay fees for layer 2 & 3 protection.
- **Result:** The buyer’s “Net Zero” claim remains intact. They do not need to scramble to buy replacement credits in a high-priced “emergency” market because the BFV math accounted for the fire risk five years before it happened.

Summary for the buyer

The BFV Framework naturally and flexibly manages their risk exposure, and accounts for the evolution of their carbon portfolio through time. In this example, regular updates of BFV rewards them for betting on an early-stage CDR developer who was able to prove out their technology and unlock additional credits for the market. At the same time, their bet on a project exposed to stochastic risks—one of the nature-based projects—was protected by BFV’s dynamic uncertainty banking approach.

Example 3: A global capital bank uses BFV to optimize for nature and net zero targets

Our final example is of a global sustainable development bank. The bank has two mandates for how it allocates capital.

- **Achieve net zero by 2050**
- **Minimize nature risks by 2030; become nature positive by 2050**

Historically, these targets have competed for financial resources. The bank has focused on engineered CDR—which typically carries no nature benefit and may present distinct risks—to reach net zero. Conversely their nature targets are met by investing in ecosystem restoration projects, whose 20-100 year time horizon is not an immediate needle-mover for their net zero targets.

The bank adopts the BFV framework and stops treating nature and net zero as separate silos. Instead, they **optimize on BFV units** which allows them to prioritize both investments simultaneously.

How the bank uses BFV to allocate capital

The bank knows that the biochar project, with its high BFV value of 0.93, would be the best bet for its net zero target. But the negative co-benefits scalar due to its use of controversial feedstocks moves their nature benchmarks in the wrong direction. They plan to put together a BFV portfolio that will allow them to balance both.

Here is their rebalancing strategy:

- **Anchoring the net zero investment:** The bank still buys 5,000 nominal tons (4,650 BFV units) from the biochar project. This significantly advances their net zero target, but also creates a “nature deficit” of 1,250 BFV units (i.e., $5,000 \times -0.25$).
- **Addressing the nature gap:** Since the mangrove project has the highest co-benefits scalar (0.4) the bank opts for a two-project strategy. The bank can use this scalar to calculate exactly the number of BFV credits it needs to purchase to close the gap created by the biochar project and remain nature positive.
- **The result:** Adding 3,125 nominal tons from the mangrove project will maintain their net zero progress while keeping the portfolio “nature neutral”. If they buy more credits from the mangrove project, or the IFM & OAE projects which also have positive (though smaller) co-benefits scalars, the portfolio becomes nature positive.

The bank has now ensured that its net zero progress remains solvent through time, but has diversified its strategy to avoid losses on the nature front. Note that, as covered in the section on dynamic uncertainty banking, the bank can also apply the carbon removal value of the mangrove project towards their net zero targets. However, they will have to regard it as a “bridge investment” towards net zero and plan to acquire more permanent removal tons in the future.

Summary for the bank

Using BFV shifts global ESG investing from a “compliance checkbox” to an actively managed **climate balance sheet**. It allows them to aggressively pursue net zero through industrial pathways while hedging their ecological impact with nature-positive credits. They are no longer guessing if they are “nature positive”—they are using metrics that quantify nature impact and incorporating these directly into their investment strategy.

Box 4: A note on fungibility of nature units

One of the principal critiques of nature markets has been that biodiversity is not exchangeable from place to place. That is to say an ecological gain at one project site cannot compensate for a loss at another, particularly if those sites occur in different biomes.

A reasonable critic could argue this applies to the co-benefits scalar encoded in the BFV. This is a valid concern, but we should emphasize that BFV is a **probabilistic accounting layer** that integrates policy standards, not a replacement for them.

Integration of co-benefits into the BFV rests on three principles:

1. **Internalizing externalities:** The co-benefits scalar is not a “nature offset” as would be widely understood by the market. Instead, it’s a recognition that a carbon project that damages surrounding ecosystems, disrupts local economies, etc. is fundamentally less solvent than a project with clear co-benefits. A BFV “fair value” ton is **risk weighted based on the project’s ecological integrity**.
2. **Accounting vs. management:** The BFV is a tool to guide capital allocation and to allow projects with different risk profiles and time horizons to be compared to one another. It is designed to work alongside standards such as TNFD (Taskforce on Nature-related Financial Disclosures) and SBTN (Science Based Targets Network) that provide guidance on how nature benefits should be managed.
3. **Systemic interdependence:** By expressing the co-benefits scalar in “carbon equivalent units” we are asserting that long-term ecological resilience is a critical goal of climate programs, alongside achieving net zero.

In short, BFV provides an actuarial framework to quantify nature benefits, and to be able to optimize portfolios on both climate and nature axes. The **ethical and ecological responsibility** for site-specific integrity remains governed by local science and global standards.

Conclusion: The Road to a Trillion-Dollar Market

Successfully executing the climate transition will require the most massive allocation of capital in human history. We simply cannot achieve it without a clear understanding of the risks embedded in the climate investments we make.

So long as we view carbon credits as a binary, where they either represent one ton value or they don’t, carbon and nature will remain fractured niche markets. High-integrity capital will simply not flow into a system that cannot mathematically define its own risks.

From nominal tons to probabilistic solvency

The Bayesian Fair Value Framework (BFV) provides the actuarial infrastructure needed to regard climate investments as probability-weighted assets. But in doing so, it is more than just an accounting framework. It is a way to align the incentives of global finance with the physical and ecological constraints that climate investments run up against.

- **For investors:** BFV provides a framework for comparing ‘fair value’ of different types of projects, and for developing risk-weighted climate portfolios.

- **For project developers:** BFV incentivizes rigorous investment, rewarding projects that reduce scientific uncertainty and execution risk, while penalizing the “junk credits” that have undermined the market.
- **For the planet:** It forces an “ecological hedge”, ensuring we don’t optimize for long-term carbon removal at the expense of the biosphere.

From carbon credits to a universal climate ledger

While the focus of this paper has been on carbon credits, BFV can be extended to include other types of climate action. Other approaches such as decarbonization, alternative environmental attribute certificates, and jurisdictional NDCs are denominated in “nominal tons”, possess varying risk profiles, and different time horizons owing to physical and policy constraints. By applying a unified “math engine” to all climate actions, we can finally compare the realized impact of an internal emission reduction against an external removal ton on an apples-to-apples basis.

Closing the integrity gap

We should not delude ourselves into believing that an accounting framework will itself be a massive demand unlock for climate. There are other structural, economic barriers that need to be addressed. However, a framework that **allows for investments to be made under clearly defined risks** is a critical prerequisite for carbon markets to scale.

This is what the BFV Framework provides. Rather than asking whether a credit is “good” or “bad”, it allows us to assess its **fair value**. This is a quantity that incorporates climate and other benefits, as well as the underlying risks inherent to projects that seek to modify a complex global carbon cycle. Under BFV, integrity is not a binary state—it is a probability distribution.

The “Fair Value” of our future depends on our willingness to price the **Cost of Certainty** today. By adopting a probabilistic accounting layer, we can finally turn the Voluntary Carbon Market into a high-functioning engine for global restoration—one that is mathematically guaranteed to deliver the future it promises.

Engagement and partnership opportunities

The BFV Framework is currently being prepared for its first cohort of pilot institutional partners. We are seeking to stress-test the model against diverse climate portfolios to refine the co-benefit scalars and uncertainty reserves. For inquiries regarding pilot participation or consulting on risk-adjusted carbon accounting, please send me a message at bclough84@gmail.com.

